

Place-Value Synthesis Review

Answer Key

Grades 4–5 Mathematics

Quick Answers

- | | |
|---------------------|--|
| 1. 62409 | 9. the total in standard form = 43070, the comparison symbol = $<$ |
| 2. 300 | 10. nearest thousand = 45000, nearest hundred = 45100, — |
| 3. $>$ | 11. the correct symbol = $<$, — |
| 4. 7400 | 12. (a) greatest to least = 42088, 41807, 41490, |
| 5. 30528 | (b) the greatest, to the nearest hundred = 42100, — |
| 6. 52000 | |
| 7. 8724, 8742, 8772 | |
| 8. 64000 | |
-

What is verified

9 of 12 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 10, 11, 12. The worked solution states what a correct response must contain.

Curriculum

Level: Grades 4–5 Mathematics

4.NBT.A.2 – *problems 1, 2, 3, 5, 7, 9, 11, 12*

4.NBT.A.3 – *problems 4, 6, 8, 10, 12*

Difficulty 1–5 of 5 across 18 tagged checks.

1. $60,000 + 2,000 + 400 + 9$ in standard form.

Solution: 6 ten-thousands, 2 thousands, 4 hundreds, 0 tens (zero holds the empty place), 9 ones. 62409

2. $8,315 = 8,000 + \square + 10 + 5$

Solution: The 8 thousands, 1 ten, and 5 ones are already listed; the digit 3 of 8,315 sits in the hundreds place, so the missing addend is 300. Check: $8,000 + 300 + 10 + 5 = 8,315$.

300

3. $36,254 \square 36,245$

Solution: Ten-thousands, thousands, hundreds all match (3, 6, 2); tens differ: $5 > 4$.

 $36,254 > 36,245$

4. Round 7,438 to the nearest hundred.

Solution: Hundreds digit 4; tens digit $3 < 5$, so round down — keep 4 hundreds.

7400

5. Which number is $30,000 + 500 + 20 + 8$?

Solution: 3 ten-thousands, 0 thousands (placeholder!), 5 hundreds, 2 tens, 8 ones.

30528

6. Round 52,499 to the nearest thousand.

Solution: Thousands digit 2; look right at the hundreds digit: $4 < 5$, so round *down*. The trailing 499 is below the midpoint 500, so $52,499 \rightarrow 52,000$.

52000

7. Order least to greatest: 8,742; 8,724; 8,772.

Solution: All share 8 thousands and 7 hundreds; compare tens: $2 < 4 < 7$.

8,724, 8,742, 8,772

8. Round 63,970 to the nearest hundred.

Solution: Hundreds digit 9; tens digit $7 \geq 5$, so round up: 639 hundreds become 640 hundreds — the carry rolls into the thousands: 64,000.

64000

9. $40,000 + 3,000 + 70$ in standard form, then compare with 43,900.

Solution: The total is 4 ten-thousands, 3 thousands, 0 hundreds (placeholder), 7 tens, 0 ones: 43,070.

43070

Compare with 43,900: hundreds differ, $0 < 9$.

43,070 < 43,900

10. 45,067 ducks: nearest thousand, nearest hundred, and which is closer?

Solution: Nearest thousand: hundreds digit $0 < 5$, round down.

45000

Nearest hundred: the hundreds block is 0 and the tens digit is $6 \geq 5$, so round up to 1 hundred:

45100

Closer: 45,100 is $45,100 - 45,067 = 33$ away, while $45,067 - 45,000 = 67$. The nearest-hundred value is closer — rounding to a smaller place always stays at least as close.

11. Ben says $76,099 > 76,105$.

Solution: Line up places. Ten-thousands, thousands match (7, 6); hundreds differ: $0 < 1$, so 76,099 is smaller.

76,099 < 76,105

Full-credit explanation: Ben judged by the eye-catching trailing nines, but the nines sit in the tens and ones places. Comparison is decided at the highest place that differs — here the hundreds — and $0 < 1$.

12. Attendances 41,807, 41,490, 42,088.

Solution: (a) Compare thousands first: 42,088 has 42 thousands, the others 41; between those, hundreds $8 > 4$.

42,088, 41,807, 41,490

(b) 42,088: tens digit $8 \geq 5$, so 0 hundreds round up to 1:

42100

(c) Check each hundreds digit against the thousand below it: 41,807 rounds up to 42,000 ($8 \geq 5$); 42,088 rounds down to 42,000 ($0 < 5$); 41,490 rounds down to 41,000 ($4 < 5$). So 41,807 and 42,088 round to the same value, 42,000 — one from below, one from above.